

Aditya Gudal

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Research engineer — robot learning, post-training, multi-agent systems.

EXPERIENCE

Nirvana Robotics — Member of Technical Staff

New York · May 2026 – present

Humanoid data pipeline (UMI, real2sim, real2real), pretraining architecture, scene-graph navigation, RL post-training. Second author on a forthcoming deformable-object manipulation benchmark with Georgia Tech.

Mbodi AI — Senior Research Engineer

USA · Sep 2025 – Jun 2026

3D reconstruction, multi-view tracking and object permanence for pick-and-place; LLM-agent control pipeline for OEM robots; open-sourced a typed robotics data library.

Nokia Bell Labs — Research Engineer

Germany & USA · Apr 2023 – Sep 2025

Decentralized energy trading (federated transformers, SSI/HSM certification, LLM-generated smart contracts); OurAI decentralized GPU marketplace; Ambion post-training platform (spun out); WhizKid multi-agent systems and failure detection; EU Horizon and U.S. government-funded distributed VLM training.

Nokia — Data Scientist

Naperville, IL · Feb 2021 – Feb 2023

First ML service and MLOps pipeline in a flagship telecom product; statistical and RL models for UE trajectory tracking, overload control and paging fan-out.

KRIBS-BioNest (RGCB / Kerala Startup Mission) — Consultant (part-time)

Kochi, India · 2018 – 2019

Genomics data science at a Government of India-established biotech incubator.

PUBLICATIONS

- AI Agent-Driven Network-Aware Decentralized Compute Resource Brokering. S. Bidkar, A. Gudal, L. Drabeck, J. Simsarian, B. Theeten, M. Khan · Optical Fiber Communication Conference (OFC) 2026, paper M3Z.15 · peer-reviewed. <https://doi.org/10.1364/OFC.2026.M3Z.15>
- CSR-RAG: An Efficient Retrieval System for Text-to-SQL at Enterprise Scale. arXiv preprint 2601.06564 · 2026. <https://arxiv.org/abs/2601.06564>
- A Ranking Framework for Network Resource Allocation and Scheduling via Hypergraphs. R. Singh, N. Boškov, A. Gudal, M. Khan · arXiv preprint 2506.01571 · 2025. <https://arxiv.org/abs/2506.01571>

PATENTS

- US 2026/0093548 A1** — Resource provider identity management, DePIN attestation and provisioning
- US 2026/0093221 A1 · EP 4723022 A1** — Renewable household energy management with federated machine learning
- US 2026/0057398 A1 · EP 4701134 A1** — Cryptographic renewable-energy verification for EV charging and grid
- Pending** — GB 2520808.3 — anomaly detection as a service for multi-agent AI systems; NC337331-IN-NP — agentic monitoring and control of an industrial network

RECOGNITION

- TechCrunch Disrupt 2025 Startup Battlefield Top 20 (Mbodi AI team)
- Automate 2026 Startup Challenge winner (Mbodi AI team)
- Nokia Bell Labs Entrepreneur-in-Residence projects: Ambion (spun out, seed-funded) and RADMA
- Seven invention disclosures at Nokia Bell Labs, three published as U.S./EU patent applications

SKILLS

Python, Rust, C++, CUDA/Triton · PyTorch, HF Transformers/PEFT/TRL, DeepSpeed, Ray · LangGraph, Pydantic AI, MCP · Docker, Kubernetes, Airflow · Open3D

EDUCATION

Northwestern University

2019 – 2020

M.S. Machine Learning & Data Science · GPA 3.9/4.0

University of Mumbai (VESIT)

2015 – 2019

B.E. Electronics Engineering · GPA 4.0/4.0